

Lesson 1: What Is AI and How Do Machines Actually Learn?

Big Idea

AI learns from lots of examples, just like you learn to recognize your friends' faces!

Learning Objectives

By the end of this lesson, students will be able to: - Explain in their own words what "AI" (Artificial Intelligence) means. - Describe the difference between a computer that follows exact RULES (like the code they wrote in Courses 10 and 11) and a computer that learns from EXAMPLES. - Name the five steps of how a simple AI model works: Examples → Training → Model → Prediction → Action. - Give at least two real-world examples of AI they already use (e.g., a phone unlocking with a face, a video app suggesting a video).

Teacher Preparation

- No website account is required for this lesson — it is a concepts + unplugged lesson.
- Optional: have a tablet/phone with a face-unlock feature or a photo-sorting app to show live (skip if unavailable — the paper activity works fully offline).
- Print or display the "Cat or Dog?" picture set from [assets/](#) (10 simple sorting cards you can sketch or print: 5 cat-like shapes, 5 dog-like shapes).
- Review the Concept Introduction below so you can explain AI simply and confidently.

Required Files

- [assets/](#) — sorting cards / picture cutouts for the Warm-up and Guided Activity.
- [Student_Lesson_Guide.md](#) , [Activities.md](#) , [Challenges.md](#) (same folder).

Teaching Flow (approx. 45 minutes)

1. Warm-up / Hook — 5 min
2. Concept Introduction — 10 min

3. Teacher Demonstration — 5 min
4. Guided Activity — 10 min
5. Mini Challenge — 5 min
6. Check for Understanding + Recap — 5 min
7. Take-Home Challenge — 5 min (assign)

Warm-up / Hook

 **Coding Mission:** Ask the class: "How do YOU know that's a dog and not a cat?" Let a few kids answer ("it has floppy ears," "it barks," "it's furry"). Then say: "You didn't read a RULE BOOK about dogs. You just SAW lots of dogs before! Today we find out computers can learn the exact same way."

Concept Introduction (keep it simple — never call it "magic")

Explain using the **Examples → Training → Model → Prediction → Action** pipeline: 1. **Examples** — We show the computer LOTS of pictures (like 50 cat photos and 50 dog photos). 2. **Training** — The computer looks at every single example, over and over, and starts noticing patterns (cats often have pointy ears, dogs often have longer noses). 3. **Model** — After training, the computer builds its own "pattern brain" called a **model**. The model is just a big set of numbers describing the patterns it noticed. 4. **Prediction** — Now show the model a NEW picture it has never seen. It uses its patterns to guess: "I think this is 90% cat." 5. **Action** — In a game or app, that guess can trigger something to happen — like a Scratch sprite saying "Meow!"

 **Robot Tip:** "I'm not magic — I'm just really, really good at spotting patterns after looking at a LOT of examples!"

 **Think Like a Coder:** Connect to prior learning — "In Course 10 and 11, YOU wrote the exact rules yourself, like `if touching edge then bounce`. Today's computer is different: instead of you writing the rule, the computer FINDS its own rule by studying examples. That's the big difference between normal code and AI."

Teacher Demonstration

1. Hold up 3 cat cards and 3 dog cards from `assets/`.
2. Say out loud the pattern you personally notice ("cats: smaller, pointier ears; dogs: bigger snouts").
3. Hold up a brand-new card face down. Ask the class to predict cat or dog based only on ears/snout — this simulates "prediction from a trained model."
4. Reveal the card and celebrate a correct group guess. 🏆 **Victory Moment!**

Guided Activity

Split into pairs. Give each pair a mixed set of 10 cards (cats/dogs, or use shapes: circles/squares if no printer). Ask them to: 1. Sort the cards into two piles (this is like collecting "Examples"). 2. Write down (or say aloud) ONE pattern rule they noticed for each pile. 3. Test their "rule" on 2 new cards you hand them at the end (this is "Prediction").

Mini Challenge

 **Game Challenge:** "Guess My Rule" — one student secretly sorts 6 objects into two groups by a hidden rule (e.g., "has straight edges" vs "does not"). Classmates guess the rule after seeing only the groups — just like a computer trying to find a pattern from examples.

Main Activity / Project

Students create a simple "AI Notebook" page (paper or digital doc) with three drawn boxes labeled **Examples**, **Model**, **Prediction**, and draw or describe one everyday example for each box (e.g., Examples = 20 photos of their own dog, Model = "knows my dog's floppy ears," Prediction = "guesses a new photo is my dog").

Fun Challenge

★ **Bonus Challenge:** Can they think of one AI they experience in real life? (music/video recommendations, spam email filters, voice assistants, face unlock). Share out loud.

Check for Understanding

Ask students to answer, thumbs up/down or verbally: - "Does a computer need to see 1 example or MANY examples to learn a pattern well?" (many) - "Is AI the same as magic?" (no — patterns from data) - "What comes after Training?" (Model)

Common Student Mistakes

- Thinking AI "already knows everything" without needing examples.
- Believing one single example is enough to train a good model.
- Confusing "the model" with "the computer itself" — clarify the model is more like a learned skill stored inside the computer.

Differentiation

- **Extra support:** Use only 2 clearly different shapes (circle vs. star) instead of cats/dogs for sorting.

- **Extra challenge:** Ask advanced students to design a sorting rule with THREE categories instead of two, and explain why more examples make the rule more reliable.

Recap

 We learned that AI is not magic — it's a computer finding patterns after studying lots of real examples, then using those patterns to make smart guesses (predictions) about new things it has never seen.

Take-Home / Creative Challenge

At home, find 5 objects in the kitchen or their room and sort them into two groups by a rule they invent. Ask a family member to guess the rule — just like a "human AI predicting a pattern."

Teacher Talking Points

- "You've been training your brain to recognize patterns your whole life — computers can now do something similar, just with numbers instead of feelings."
- "Remember: more good examples = a smarter, more reliable model. Few or messy examples = a confused model."
- "Next lesson, YOU get to train a real AI model yourselves using your webcam!"

Fun elements used in this lesson:  Coding Mission ·  Robot Tip ·  Think Like a Coder ·  Bonus Challenge ·  Game Challenge ·  Debug Detective ·  Make It Yours ·  Victory Moment